



Bangladesh University of Business and Technology

Department of Computer Science and Engineering

CSE 498A / CSE 498B: Capstone Project Progress Journal

CAPSTONE PROJECT TITLE: A Low-Cost Vision-Based Autonomous Navigation System for Mobile Robots Using Deep Learning and Geometric-AI Perception

SUPERVISOR NAME: Md. Mahbub-Or-Rashid

COURSE TEACHER: Dr. Muhammad Aminur Rahaman

FULL ID	FULL NAME	INTAKE/SEC
22235103242	Tarikul Islam	51/05
22235103227	Sultan Ahmmed	51/05
22235103247	Md Soykot Ahamed	51/05
22235103237	Ali Akbar Rakib	51/05
22235103235	Suraia Islam	51/05

SUPERVISOR & COURSE TEACHER APPROVAL

Supervisor's Signature with date

Course Teacher's Signature with date



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Capstone Project Progress Entry

Entry Date: 1/3/2026

Previous Entry Date: N/A

Previous Issues Discussed: N/A

Current Research Issue: Identifying a practical and low-cost research area for autonomous navigation.

Research Goal for this Session: To explore possible project domains and decide on a suitable direction.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Reviewed basic robotics and vision-based navigation concepts	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Compared several obstacle avoidance approaches	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Studied monocular camera-based systems	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Observed real-world autonomous robot use cases	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Summarized possible project directions	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: The team discussed different robotics and computer vision ideas and identified obstacle avoidance using monocular vision as a promising and affordable research direction.

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Entry Date: 1/15/2026

Previous Entry Date: 1/3/2026

Previous Issues Discussed: Project direction and initial feasibility study.

Current Research Issue: Comparing similar research works and identifying key challenges.

Research Goal for this Session: To analyze existing solutions and determine the main research gaps.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Read foundational papers on robotic obstacle avoidance	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Identified common problems in vision-based navigation	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Compared sensor-based and camera-based methods	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Listed engineering constraints for mobile robots	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Prepared notes on research gaps	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: Several related papers were reviewed to understand the problem more clearly. The team noticed that camera-based navigation can reduce cost, but accurate obstacle estimation remains a major challenge.

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Capstone Project Progress Entry

Entry Date: 1/28/2026

Previous Entry Date: 1/15/2026

Previous Issues Discussed: Research gaps and existing approaches.

Current Research Issue: Finalizing the topic and shaping the project title.

Research Goal for this Session: To select a final research theme and refine the project title.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Narrowed down the project scope	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Discussed title alternatives	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Evaluated title feasibility	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Checked technical relevance	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Finalized project title draft	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: The final title was selected after discussion, keeping in mind cost, feasibility, and research value.

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Capstone Project Progress Entry

Entry Date: 1/12/2026

Previous Entry Date: 1/28/2026

Previous Issues Discussed: Final title selection and scope narrowing.

Current Research Issue: Drafting the proposed methodology and system flow.

Research Goal for this Session: To design the methodology structure and plan the implementation pipeline.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Drafted the overall system workflow	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Organized the problem statement	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Prepared the methodology outline	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Matched tasks with research objectives	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Reviewed implementation feasibility	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: A preliminary methodology was developed, showing how monocular vision and geometric reasoning can be combined for obstacle avoidance.

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Capstone Project Progress Entry

Entry Date: 3/24/2026

Previous Entry Date: 1/12/2026

Previous Issues Discussed: Methodology planning and system workflow design.

Current Research Issue: Preparing the proposal document and supporting analysis.

Research Goal for this Session: To complete the research proposal with objectives, methodology, and expected contribution.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Wrote the problem statement	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Drafted the objectives	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Summarized related work	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Prepared the methodology section	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Finalized proposal content	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: *The proposal was completed with the necessary sections and prepared for submission.*

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Entry Date: 4/7/2026

Previous Entry Date: 3/24/2026

Previous Issues Discussed: Proposal preparation and methodology planning.

Current Research Issue: Final review and submission of the capstone proposal.

Research Goal for this Session: To revise the document and submit the final proposal package.

FULL ID	ACTIVITIES	RESULTS	EVALUATION (Click on the Checkbox)
22235103242	Checked the full proposal draft	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103247	Reviewed formatting and references	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103227	Verified technical sections	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103237	Prepared submission materials	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory
22235103235	Submitted the project proposal	Done	☐ Satisfactory ☐ Need Improvement ☐ Unsatisfactory

Remarks: *The proposal was completed with the necessary sections and prepared for submission.*

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